

An Automated Platform for GNSS Processing and Time-Series Analysis in a Complex Geodynamic Region

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Outline

- GNSS Networks in Greece
- Processing and Analysis Platform Upgrade
- Results & Outputs
- Time series analysis results
- Discussion / Conclusions



Motivation

Geodynamic context

Greece is one of the most active geodynamic regions in Europe, affected by tectonic deformation, strong earthquakes, and volcanic activity, such as in Santorini.

Routine GNSS processing and site/network monitoring allow us to:

- produce reliable and repeatable geodetic products for geodesy, seismology, volcanology, and broader Geoscience applications;
- estimate long-term station velocities and detect transient deformation, offsets, and post-seismic or volcanic signals;
- maintain processing strategies consistent with modern GNSS standards, reference frames, and EUREF/IGS practices;
- develop an automated platform for rapid, final, and high-rate GNSS solutions, supporting both research and academic services.

Key idea

Routine processing transforms heterogeneous GNSS observations into consistent time series and deformation products for a complex geodynamic region.

The Data Set and metadata challenges

Routine processing for precise positioning requires a well-established and reliable data set with complete metadata.

Currently we process whatever we can get our hands on ...

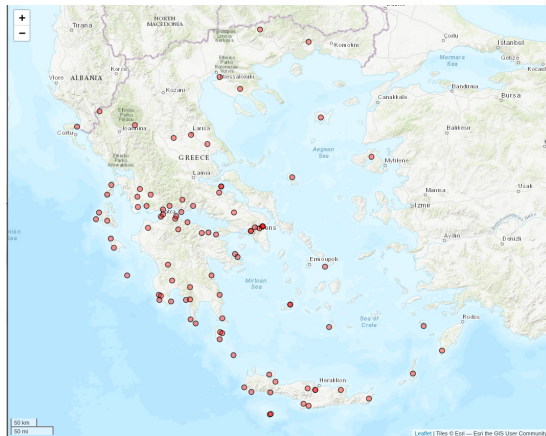
Problems:

- Inhomogeneous dataset (RINEX of various versions, raw files, etc).
- Various maintainers, different mentalities.
- Different acquisition methods/rates.
- No log files for maintainers with no geodetic interest (e.g. surveying companies).
- Wide variety of equipment (not always included in ATX files).

Network GREECE

Network Greece includes the majority of the available sites (≈ 100) but not all of them are (always/currently) active. Various providers but all with geodetic interest & equipment.

- covers all of Greece
- different (geodetic type) equipment
- credible time-span (early 2004 - now)
- all freely available GNSS data
- large data gaps & inactive stations

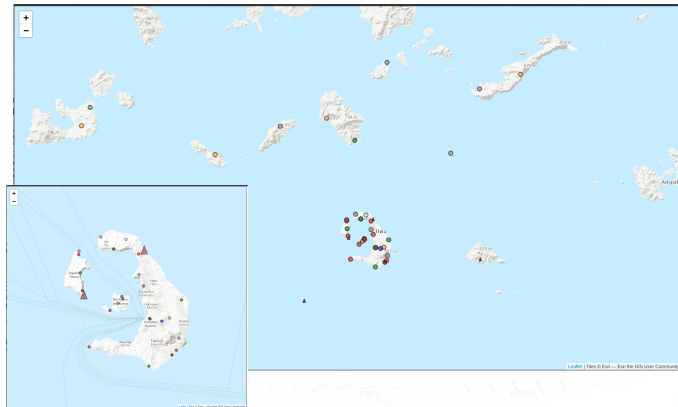


-> Network GREECE

Network SANTORINI

The Santorini Volcano ...

- credible time-span
- only covers the wider area of Santorini Volcano
- different providers (including surveying & cadastral services)
- no log files & equipment changes



-> Network Santorini

Network HEPOS

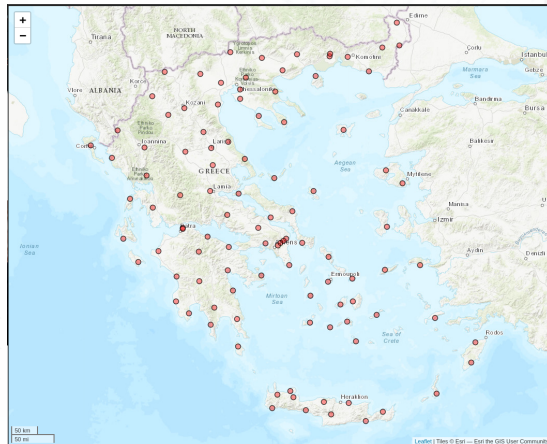
Network HEPOS

- covers all of Greece
- homogeneous equipment
- credible time-span (mid 2007 - now)
- only four years available for processing up to now

*Data availability and
funding from
Hellenic Cadastre*



HELLENIC CADASTRE



-» Network HEPOS

GNSS data processing and analysis at DSO

GNSS data analysis

- Bernese v5.4 ([Dach, 2015](#))
- PRIDE-PPPAR ([Geng et al., 2019](#))
- GipsyX ([Bertiger et al., 2020](#))

Time series analysis

- Hector v2.1 ([Bos et al., 2012](#))
- Python/C++ routines

Solutions

- Daily solution
 - Rapid solution - 1 day after
 - Final solution - 12 days after
- High-rate analysis (>1 Hz)
 - data from network HEPOS - 1 Hz



Automation Workflow

The core tool/software is **Bernese GNSS Software v5.4**.

Integration with

- **PostgreSQL** database,
- **Python** module (product/data downloading, pre-processing, driving cron jobs, etc)
<https://github.com/DSOlab/autobern>
- **Time-series** analysis (integrated in routine processing on regular intervals)
- **Strain Rates** via StrainTool (on user demand)

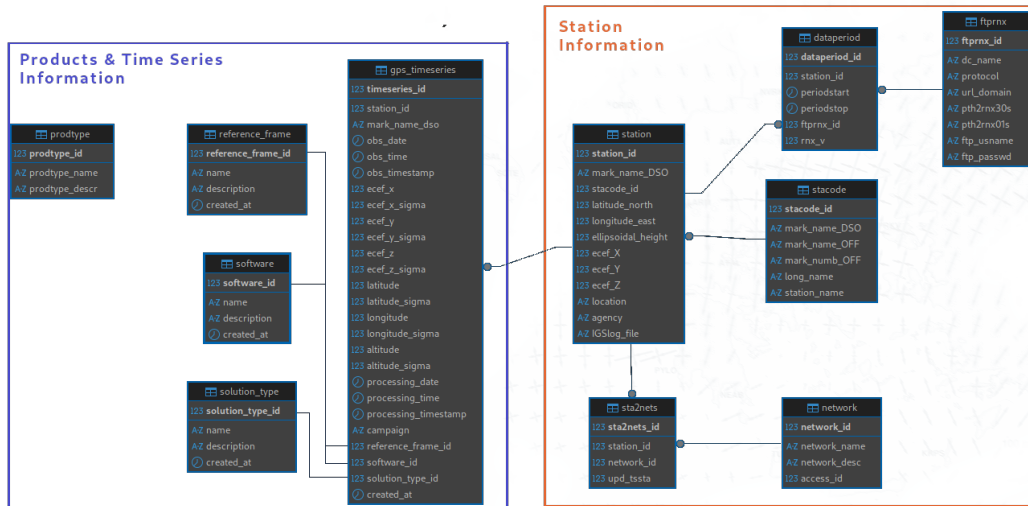


IGS20 transition for REPRO2026

- Reference frame [IGS20](#) > [IGb20](#) > [IGc20](#)
- Oceans loading corrections ([NTUA_FES2014b](#))
- [VMF3](#) for tropospheric modeling
- ATX file: [igs20.atx](#)
- Long filename of products according to new IGS convention

	Server	Start	Finish	REVIEW	UPLOAD EPN	start day	proc days	end day	Last script run	COMMENTS
1995										
1996										
***	***	***	***	***	***	***	***	***	***	***
2018										week=2008 start BIA, jul 1st/182
2019										
2020	bernese1	13.06.2026		ongoing		001	224	366	Wed Jun 17 21:34:48 UTC 2026	
2021	paraskevas	12.06.2026		ongoing		001	265	365	Wed 17 Jun 08:29:09 EEST 2026	
2022	dso01	12.06.2026		ongoing		001	262	365	Wed Jun 17 08:32:43 EEST 2026	2022-331 -> change to new long
2023	bernese1	05.06.2026	12.06.2026	4REVIEW	OK ALL	001		365	FINISHED	
2024	paraskevas	02.06.2026	11.06.2026	4REVIEW	OK ALL	001		366	FINISHED	
2025A	dso01	02.06.2026	04.06.2026	4REVIEW	OK ALL	001		050	FINISHED	
2025B	dso01	06.06.2026	11.06.2026	4REVIEW	OK ALL	051		365	FINISHED	switch to IGB20 02.02.2025
2026	dso01					033		050		USE IGS20 check the results
2026B	vincenty	16.06.2026		4REVIEW		001		010	FINISHED	
2026C	vincenty	16.06.2026		ongoing		011	075	147	Tue Jun 16 11:54:16 EEST 2026	switch to IGC20 11.01.2026
2027										

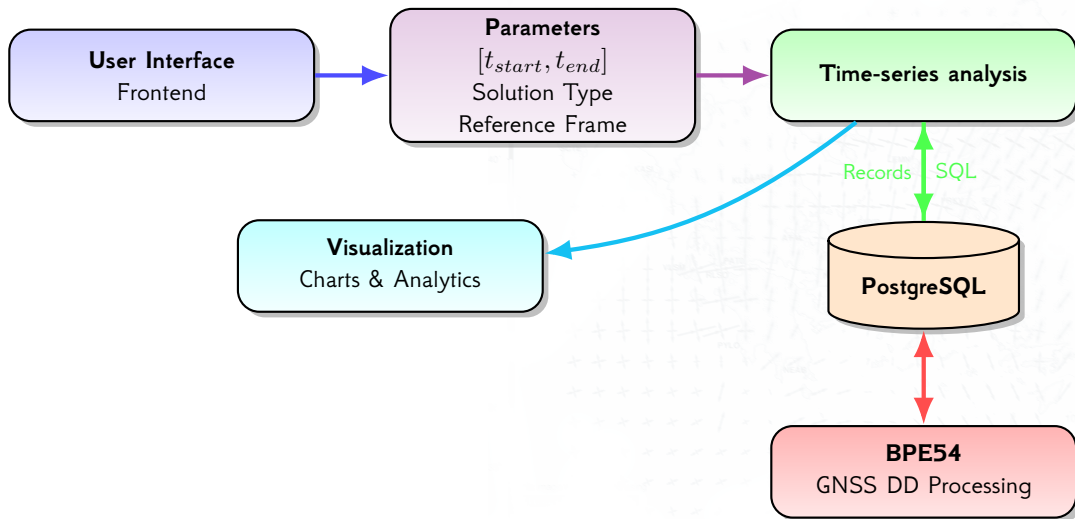
Transition to the New PostgreSQL Database Schema



Time-series analysis ⇐

⇒ BPE54

Platform Workflow



Web Interface: Station Selection and Filtering

- User entry point to the application.
- Selection of basic parameters:
 - ◇ GNSS station
 - ◇ Solution type
 - ◇ Software
 - ◇ Reference frame
- Filtering and reordering capability

Live app (beta version):



GPS Station List

Network Coverage Map Hide Map

Search by station code, location, or name

Solution Type:

☐ Final (120,425)

☐ Rapid

☐ Ultra_Rapid (7,714)

☐ Broadcast

Software:

BERNESE52 (128,139) ▼

Reference Frame:

All ▼

Apply Filters

Clear All

Total Stations: 207 Stations Shown: 207 Network: Greece GNSS

Total stations: 207

Station Code	Official Name	Location	Network	DOMES Number	Records	First Date	Last Date	Actions
vts1	vts1	Voutes, Heraklio, GR	greece	-	402	2025-06-05	2026-02-20	View Charts
irkl	irkl	Iraklia	greece	-	503	2025-04-10	2026-02-20	View Charts
kori	kori	Korinthos, GR	greece	-	675	2022-02-04	2026-02-20	View Charts
vism	vism	Valsamata	greece	12625M001	2814	2008-01-01	2026-02-20	View Charts

<http://83.212.80.177:8585/>

Processing Outputs and Products

4. Solution Indetifiers

Array of Objects

expand

5. PCF Variables

Array of Objects

expand

6. Saved products

Array of Objects

expand

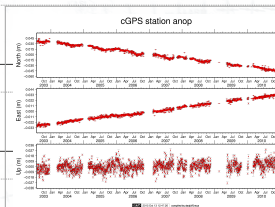
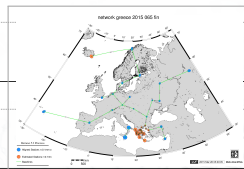
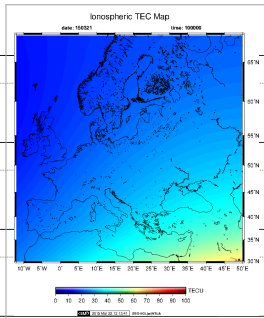
7. Warnings

Array of Objects

expand

8. Ambiguity Resolution Summary

Array of Objects

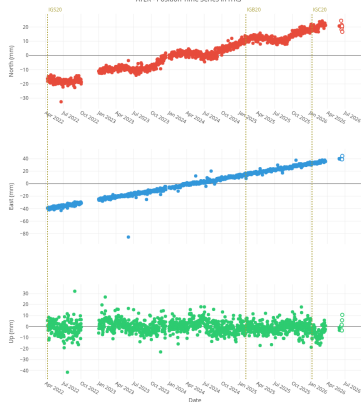


Baseline	sta1	sta2	length (km)	Method	N. of Amb.	Percentage	Satellite system
AUKL	AUT1	KLOK	139.7	pbnl	74	54.1	GPS
AULE	AUT1	LEMN	199.6	pbnl	60	55	GPS
KCTL	KATC	TILO	59	pbnl	50	90	GPS
KLRL	KLOK	RLSO	174.2	pbnl	74	41.9	GPS

Online Time-Series Products

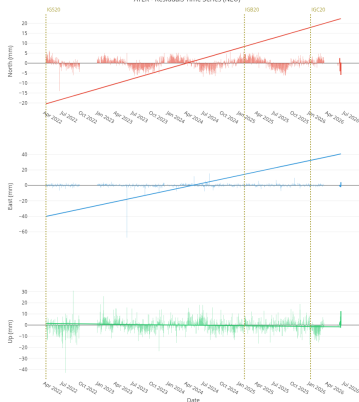
Times-series

ATER - Position Time Series in ITRS



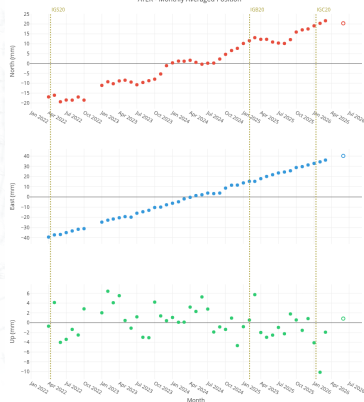
Linear regression residuals

ATER - Residuals Time Series (NEU)



Monthly average

ATER - Monthly Averaged Position



Time series Model

Linear+Seasonal+PSD Terms

$$y(\delta t) = A\delta t + B + C_1 \sin(\omega_1 \delta t) + D_1 \cos(\omega_1 \delta t) + C_2 \sin(\omega_2 \delta t) + D_2 \cos(\omega_2 \delta t) \\ + \sum_{k=1}^n J_k H(t - t_k) + \sum_{i=1}^n A'_i \log\left(1 + \frac{\delta t}{T'_i}\right) + \sum_{j=1}^n A_j \left(1 - e^{-\frac{\delta t}{\tau_j}}\right)$$

$\delta\tau$ ($= \tau - \tau_0$): time elapsed since the central epoch τ_0 [yr]

A : station velocity [m/yr]

B : position intercept at τ_0 [m]

C_k, D_k : sine and cosine amplitudes at angular frequency ω_k [m], with $\omega_1 = 2\text{rad/yr}$ (annual) and $\omega_2 = 4\text{rad/yr}$ (semi-annual)

J_k : magnitude of the k-th step offset [m], modelled by the Heaviside function $H(\tau - \tau_k)$ activating at epoch τ_k ; v is the total number of offsets

A'_i, A_i^e : amplitudes of the i-th logarithmic and exponential PSD terms [m]

τ_I^l, τ_i^e : relaxation time constants of the i-th logarithmic and exponential PSD terms [yr]

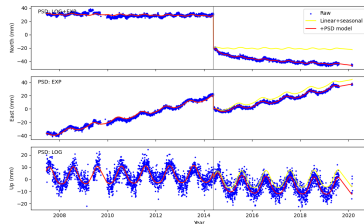
Δt_i ($= t - t_i^e$ if $t > t_i^e$): time elapsed since the i-th earthquake epoch t_i^e ; all PSD terms are zero before the earthquake

n, n : number of logarithmic and exponential PSD terms

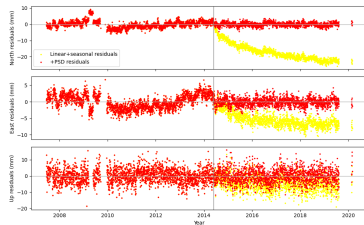
Impact of PSD Modelling on Greek GNSS Time Series

Preliminary results

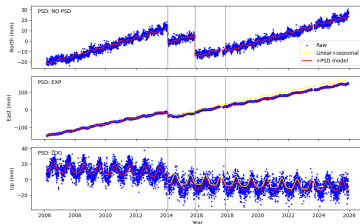
LEMN.txyz2 trajectory



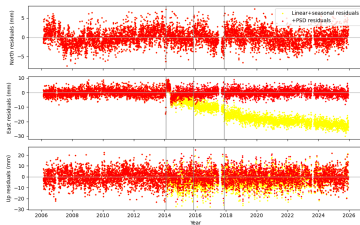
LEMN.txyz2 residuals



VLSM.txyz2 trajectory

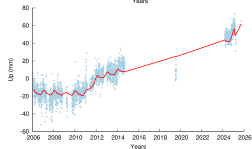
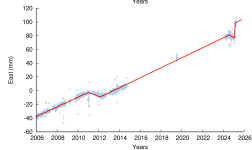
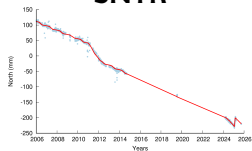
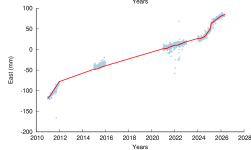
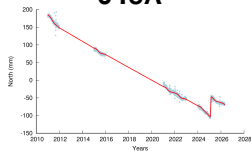
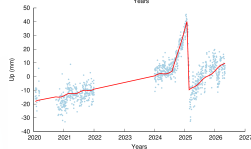
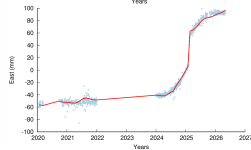
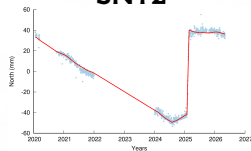
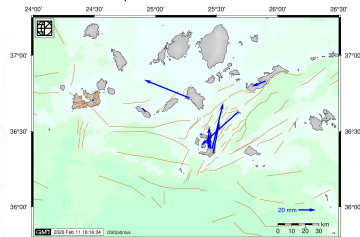
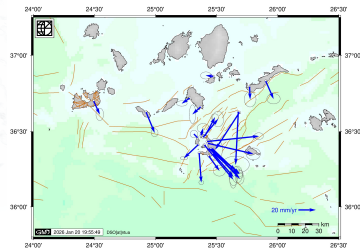


VLSM.txyz2 residuals



[ToDo] Differences in PSd parameters....

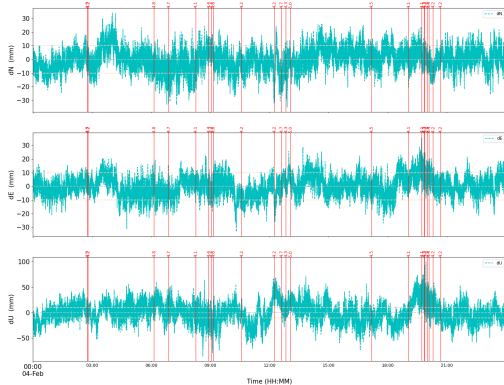
Complex station displacements - Santorini Island

SNTR**048A****SNT2****Horizontal displacements 20250201 - 20250228 [IGS20]****Horizontal velocities 20250303 - 20250808 [IGS20]**

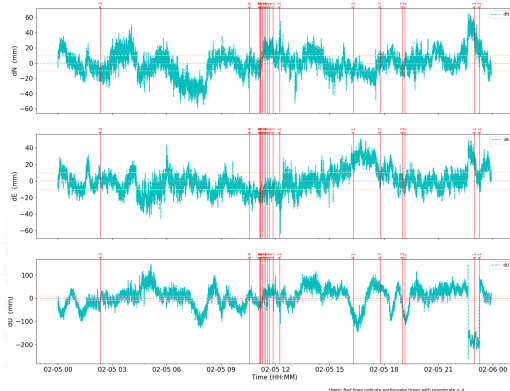
PPP processing for High-rate data

Santorini Volcano crisis (02.2025)

048A_2025_035



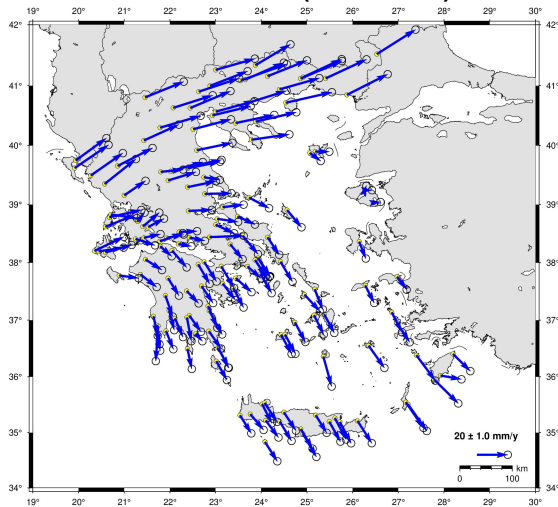
048A_2025_036



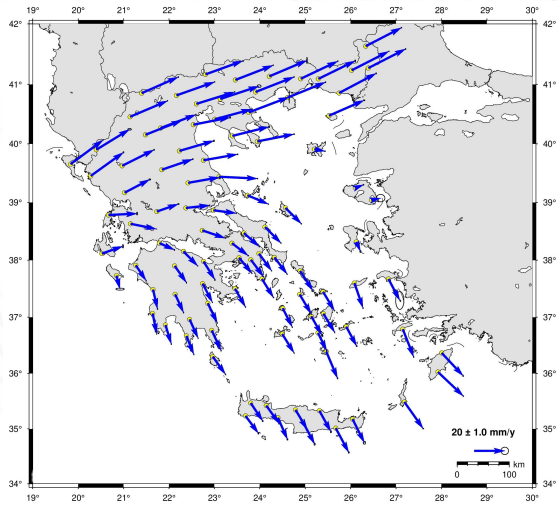
- 1 Hz data available, ~5 stations from HEPOS Network.
- PPP-AR analysis for 2 days (DOY: 035, 036)

REPRO2026: Preliminary Velocity Fields and Network Comparison

DSO_GRC_2026(2020-2026)



DSO_HPS_2023



Conclusions and Future work

- Greece's complex tectonic and volcanic setting requires continuous and reliable GNSS monitoring.
- Routine GNSS processing provides consistent time series, station velocities, and deformation products for Geoscience applications.
- The upgraded DSO platform automates data handling, processing, database storage, time-series analysis, and visualization.
- Including post-seismic deformation (PSD) terms in time-series analysis is essential to separate transient signals from long-term tectonic velocities and avoid biased deformation estimates.
- Dense GNSS networks improve the realization of a stable local reference frame and help detect earthquakes, volcanic signals, and other transient phenomena.
- Future work focuses on expanding the datasets, supporting rapid and high-rate solutions, and strengthening DSO's contribution to the GNSS/EUREF community.

Thank you for your attention! 🇸🇦🇩🇪



References I



Bertiger, Willy et al. (Aug. 2020). "GipsyX/RTGx, a new tool set for space geodetic operations and research". en. In: *Advances in Space Research* 66.3, pp. 469–489. ISSN: 02731177. DOI: [10.1016/j.asr.2020.04.015](https://doi.org/10.1016/j.asr.2020.04.015). URL: <https://linkinghub.elsevier.com/retrieve/pii/S0273117720302532> (visited on 06/18/2026) (cit. on p. 8).



Bos, M. S., R. M. S. Fernandes, S. D. P. Williams, and L. Bastos (Dec. 2012). "Fast error analysis of continuous GNSS observations with missing data". In: *Journal of Geodesy* 87.4, pp. 351–360. DOI: [10.1007/s00190-012-0605-0](https://doi.org/10.1007/s00190-012-0605-0) (cit. on p. 8).



Dach, Rolf (2015). "Bernese GNSS Software Version 5.2, User Manual". In: ed. by Rolf Dach, Simon Lutz, Peter Walser, and Pierre Fridez. Astronomical Institute, University of Bern, Bern Open Publishing. ISBN: 978-3-906813-05-9 (cit. on p. 8).



Geng, Jianghui, Xingyu Chen, Yuanxin Pan, Shuyin Mao, Chenghong Li, Jinning Zhou, and Kunlun Zhang (Oct. 2019). "PRIDE PPP-AR: an open-source software for GPS PPP ambiguity resolution". en. In: *GPS Solutions* 23.4, p. 91. ISSN: 1080-5370, 1521-1886. DOI: [10.1007/s10291-019-0888-1](https://doi.org/10.1007/s10291-019-0888-1). URL: <https://link.springer.com/10.1007/s10291-019-0888-1> (visited on 06/18/2026) (cit. on p. 8).